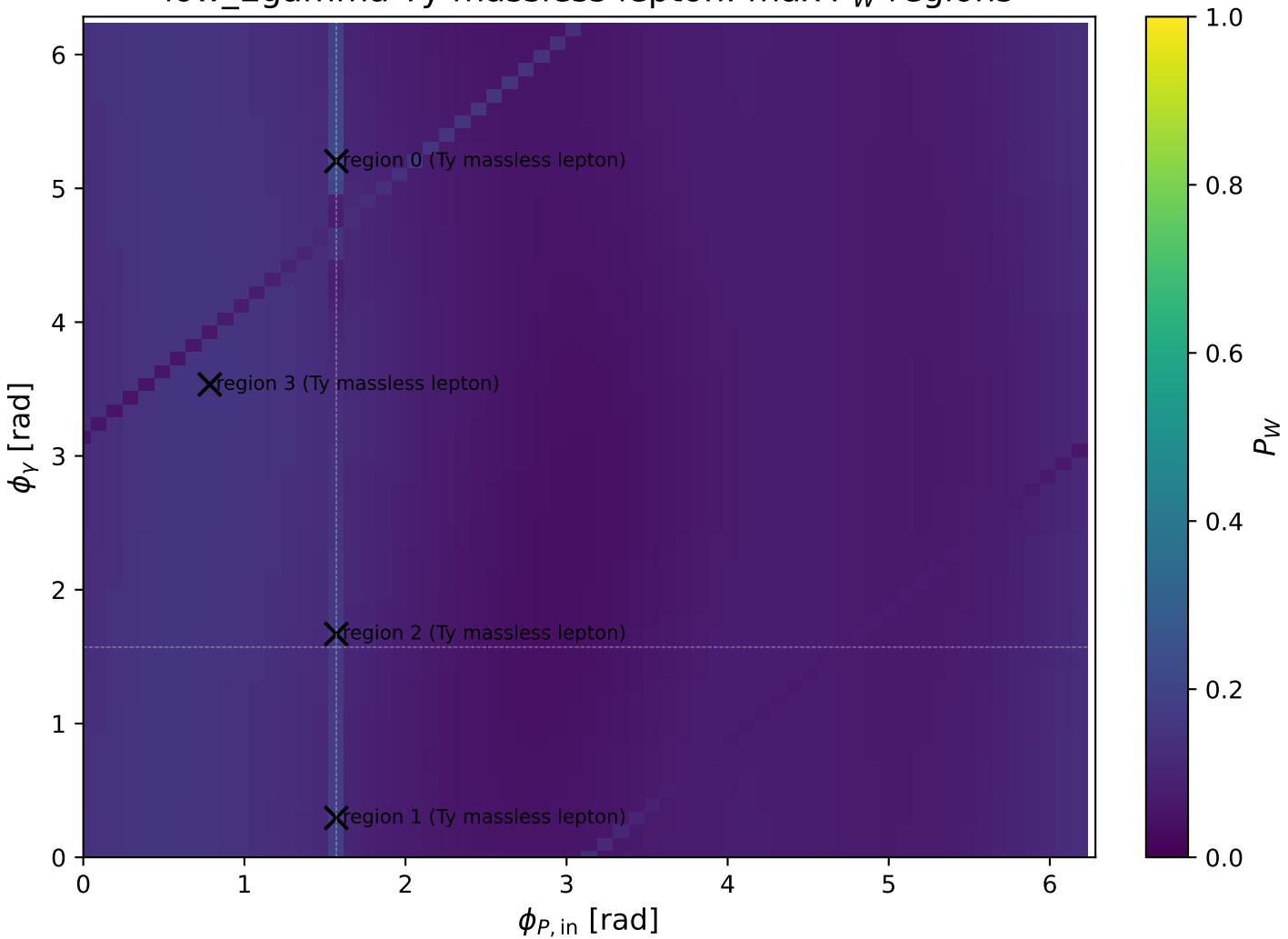
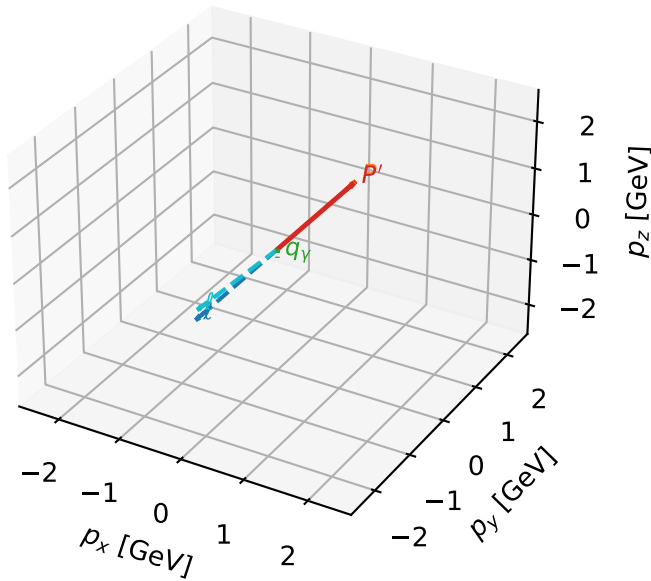


low_Egamma Ty massless lepton: max P_W regions



Ty massless lepton characteristic max P_W configuration

3D momenta

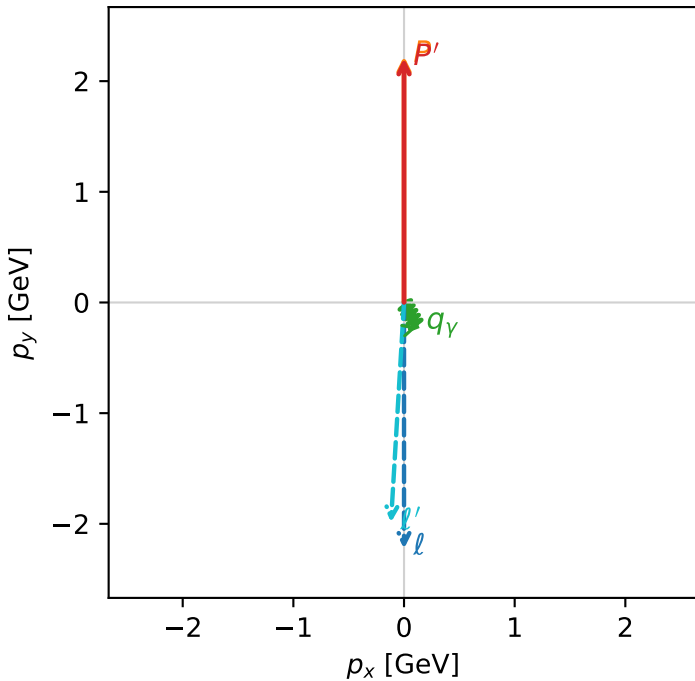


w_purity low egamma ty lepton_region_0 P_W=0.229452
 region: low_Egamma_Ty_lepton_region_0
 outgoing-state purity: 0.5
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(T_{Y, l_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.20723$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_P=1.5708$
 $E_Y=0.25$, $\phi_Y=5.20326$
 $Q^2=0.0155362$, $x_B=0.00710071$, $t=-4.48838e-05$, $W^2=3.05228$, $y=0.106027$
 $\theta(l', \gamma)=31.5197$ deg

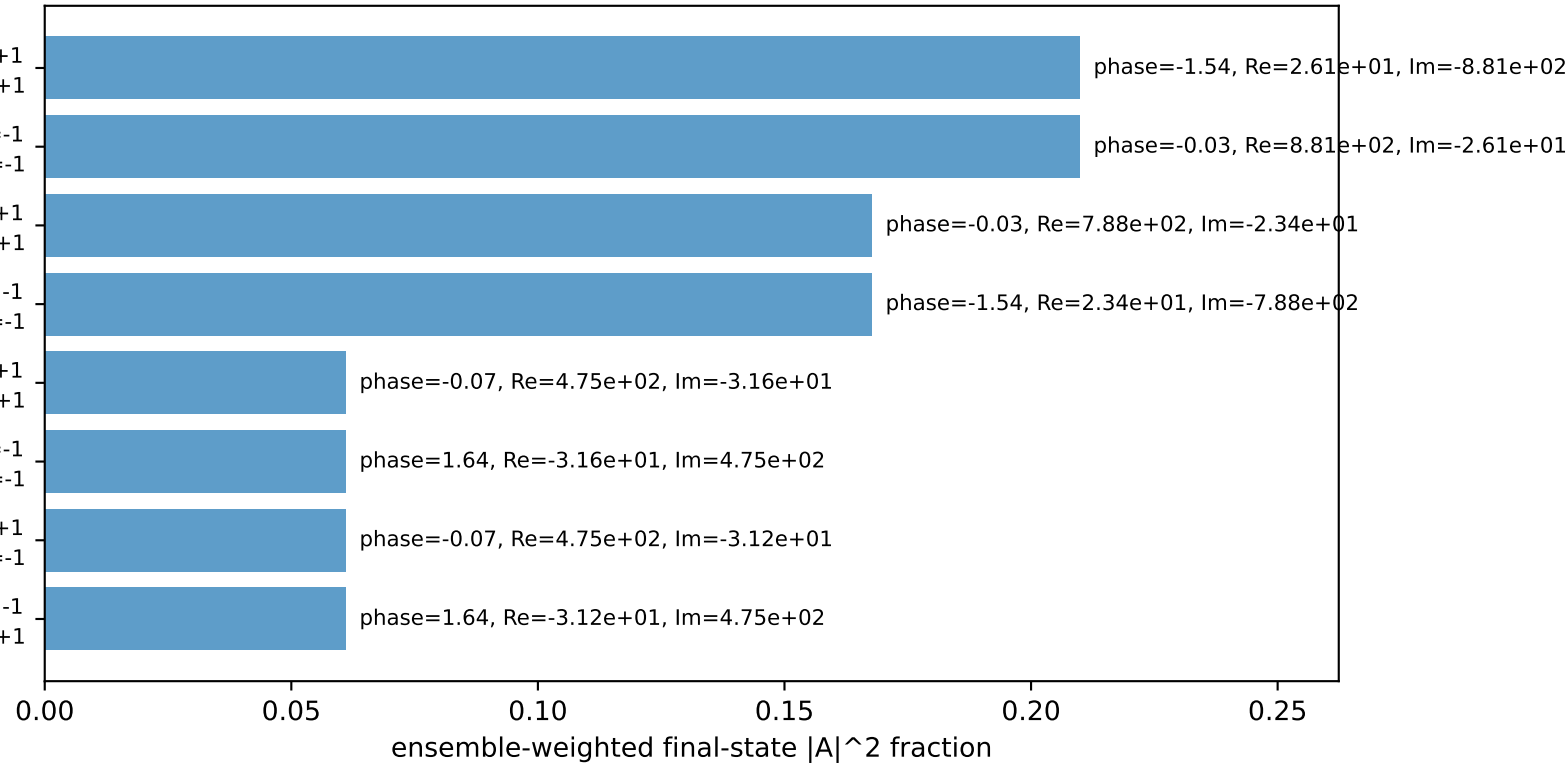
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 l' $E= 1.9902$ $p_x= -0.1178$ $p_y= -1.9868$ $p_z= 0.0000$
 P' $E= 2.3983$ $p_x= 0.0000$ $p_y= 2.2072$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.1178$ $p_y= -0.2205$ $p_z= 0.0000$

Transverse plane



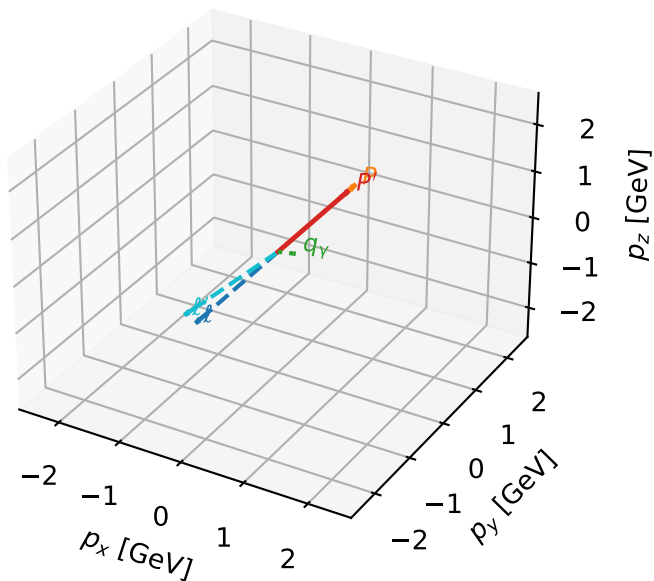
$h_e=+1, h_p=-1, h_Y=+1$ electron Ty, proton h=+1	phase=-1.54, Re=2.61e+01, Im=-8.81e+02
$h_e=-1, h_p=+1, h_Y=-1$ electron Ty, proton h=-1	phase=-0.03, Re=8.81e+02, Im=-2.61e+01
$h_e=-1, h_p=-1, h_Y=+1$ electron Ty, proton h=+1	phase=-0.03, Re=7.88e+02, Im=-2.34e+01
$h_e=+1, h_p=+1, h_Y=-1$ electron Ty, proton h=-1	phase=-1.54, Re=2.34e+01, Im=-7.88e+02
$h_e=-1, h_p=+1, h_Y=+1$ electron Ty, proton h=+1	phase=-0.07, Re=4.75e+02, Im=-3.16e+01
$h_e=+1, h_p=-1, h_Y=-1$ electron Ty, proton h=-1	phase=1.64, Re=-3.16e+01, Im=4.75e+02
$h_e=-1, h_p=-1, h_Y=+1$ electron Ty, proton h=-1	phase=-0.07, Re=4.75e+02, Im=-3.12e+01
$h_e=+1, h_p=+1, h_Y=-1$ electron Ty, proton h=+1	phase=1.64, Re=-3.12e+01, Im=4.75e+02

Leading initial-ensemble/final-state components (N=8, retained=100.0%)



Ty massless lepton characteristic max P_W configuration

3D momenta

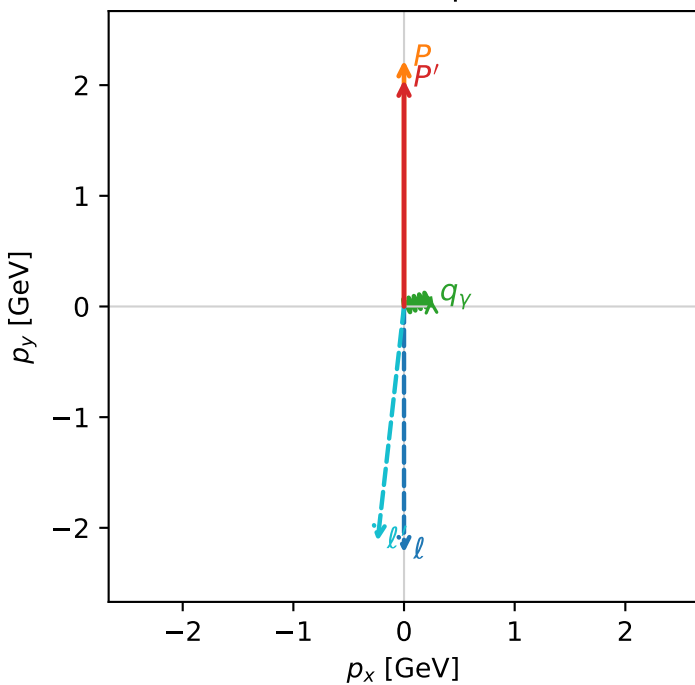


w_purity low egamma ty lepton_region_1 P_W=0.182798
 region: low_Egamma_Ty_lepton_region_1
 outgoing-state purity: 0.5
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(T_{y, l_0}, h_p)$

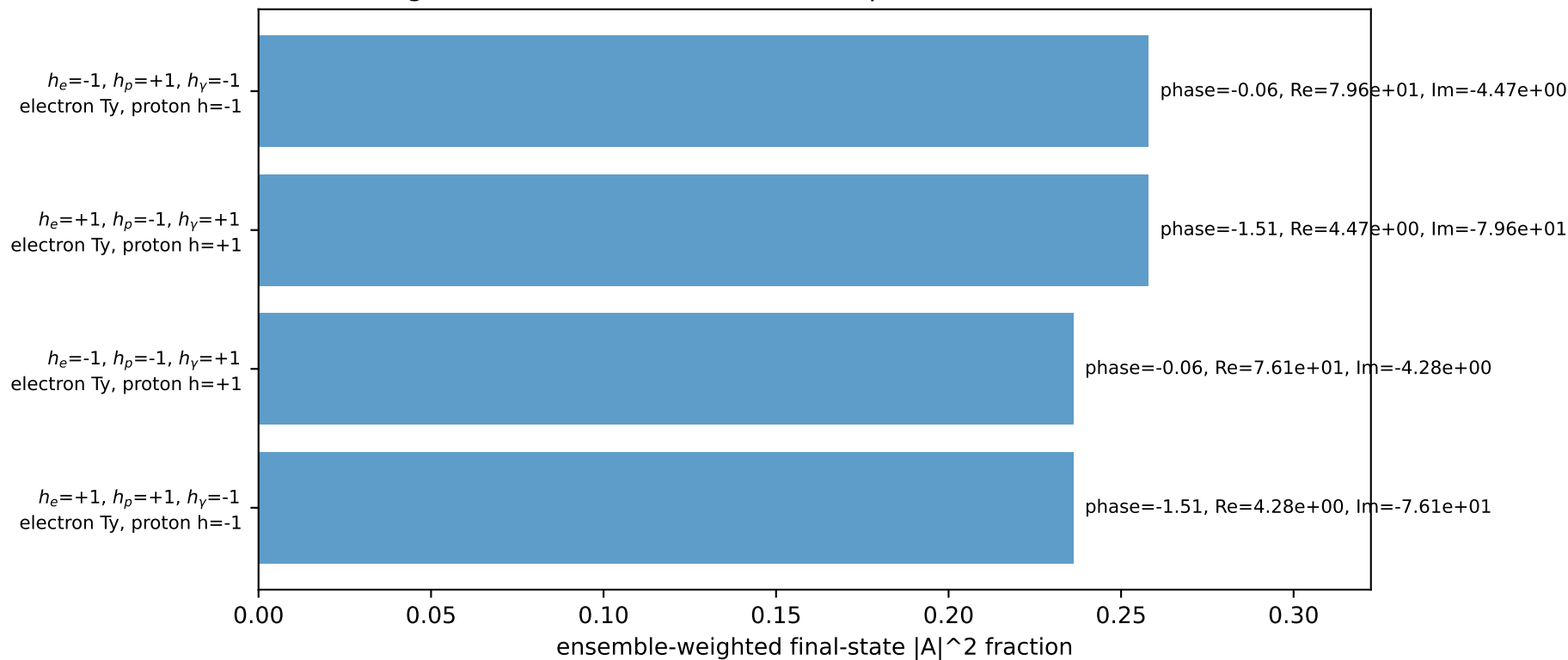
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.049$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_P=1.5708$
 $E_Y=0.25$, $\phi_Y=0.294524$
 $Q^2=0.0598183$, $x_B=0.0672743$, $t=-0.00497769$, $W^2=1.7092$, $y=0.0430883$
 $\theta(l', \gamma)=113.309$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 l' $E= 2.1350$ $p_x= -0.2392$ $p_y= -2.1216$ $p_z= 0.0000$
 P' $E= 2.2535$ $p_x= 0.0000$ $p_y= 2.0490$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.2392$ $p_y= 0.0726$ $p_z= 0.0000$

Transverse plane

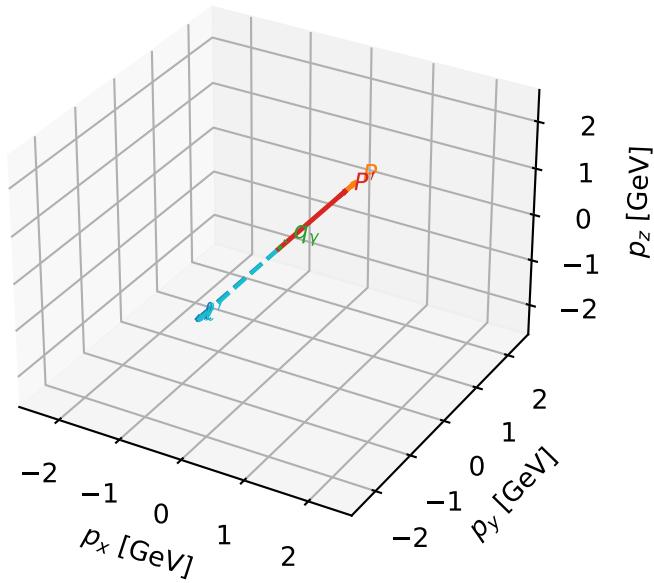


Leading initial-ensemble/final-state components (N=4, retained=98.8%)



Ty massless lepton characteristic max P_W configuration

3D momenta

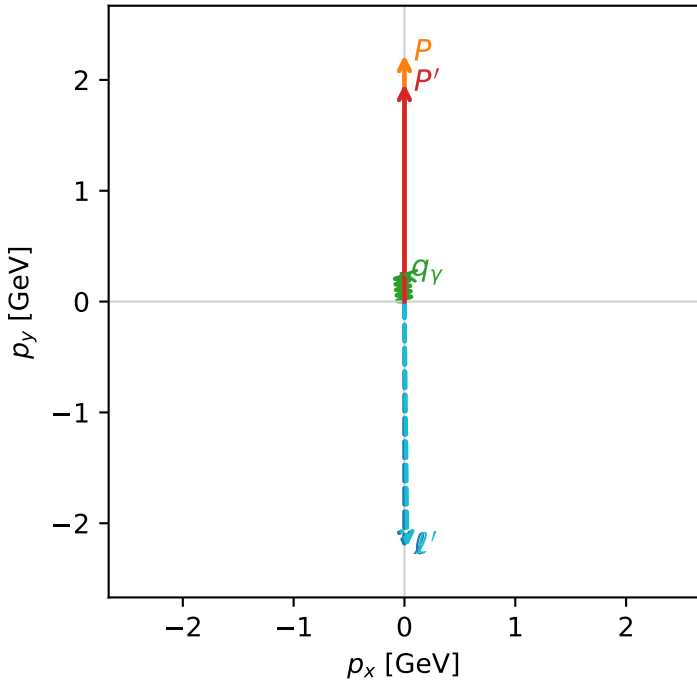


w_purity_low_egamma_ty_lepton_region_2 P_W=0.165397
 region: low_Egamma_Ty_lepton_region_2
 outgoing-state purity: 0.5
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(T_{Y,\ell_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.96352$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_P=1.5708$
 $E_Y=0.25$, $\phi_Y=1.66897$
 $Q^2=0.000603726$, $x_B=0.00540958$, $t=-0.0114065$, $W^2=0.990843$, $y=0.00540818$
 $\theta(\ell', \gamma)=175.01$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 2.2125$ $p_x= 0.0245$ $p_y= -2.2123$ $p_z= 0.0000$
 P' $E= 2.1761$ $p_x= 0.0000$ $p_y= 1.9635$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.0245$ $p_y= 0.2488$ $p_z= 0.0000$

Transverse plane



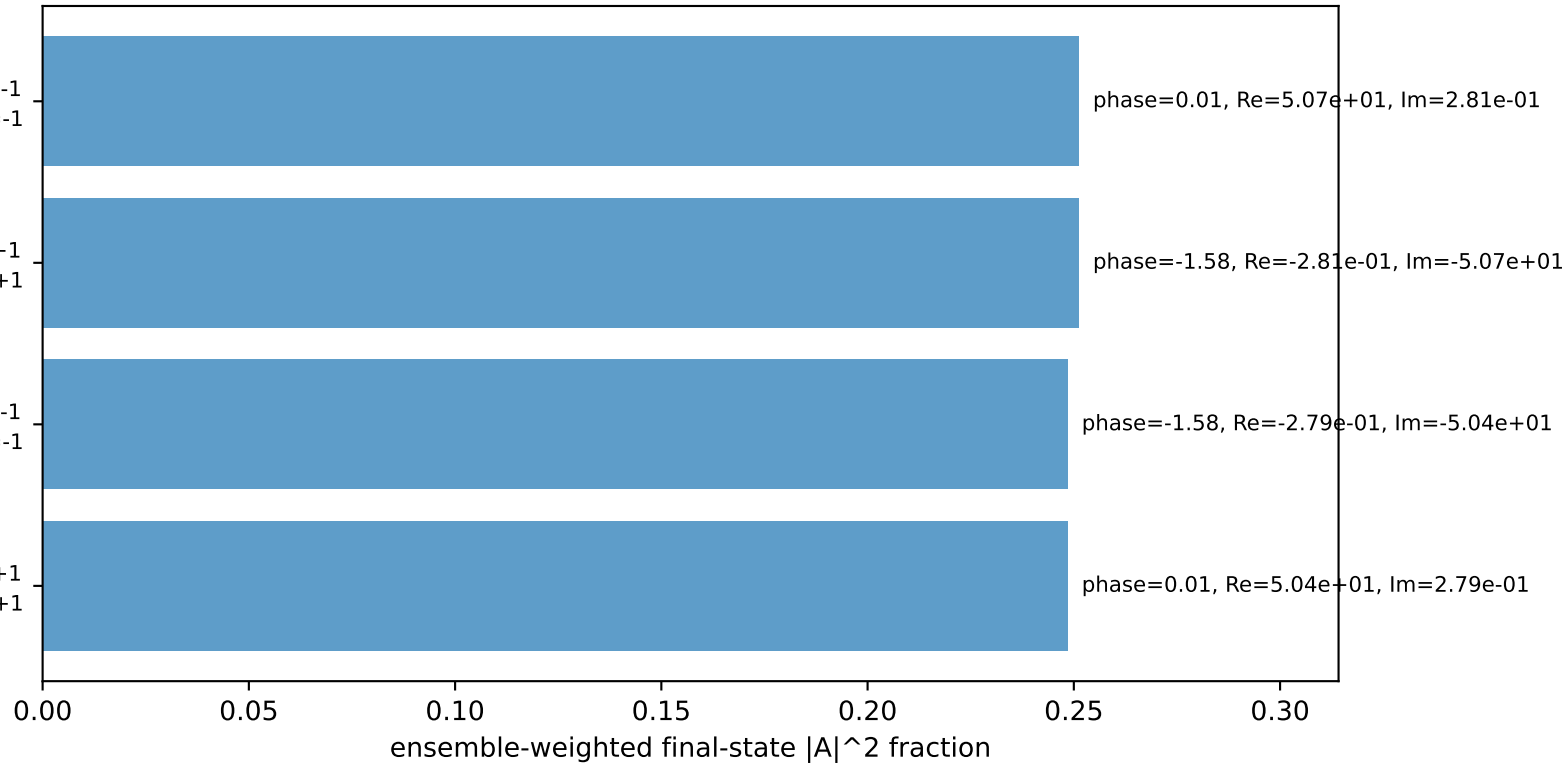
$h_e=-1, h_p=+1, h_Y=-1$
 electron Ty, proton h=-1

$h_e=+1, h_p=-1, h_Y=+1$
 electron Ty, proton h=+1

$h_e=+1, h_p=+1, h_Y=-1$
 electron Ty, proton h=-1

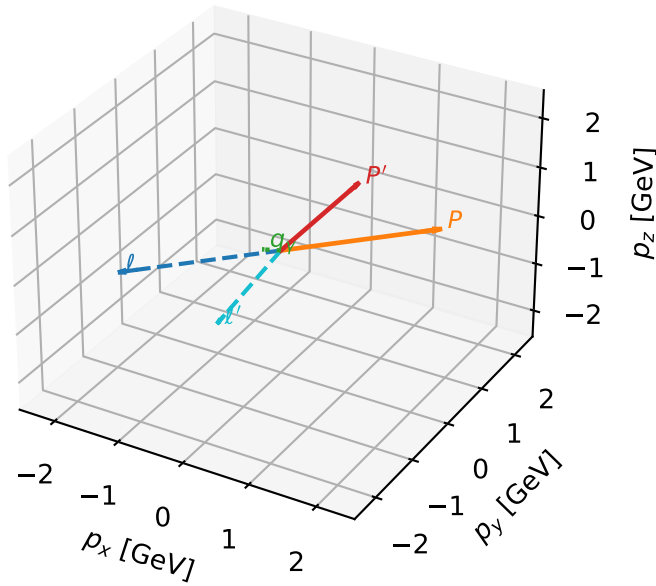
$h_e=-1, h_p=-1, h_Y=+1$
 electron Ty, proton h=+1

Leading initial-ensemble/final-state components (N=4, retained=100.0%)



Ty massless lepton characteristic max P_W configuration

3D momenta



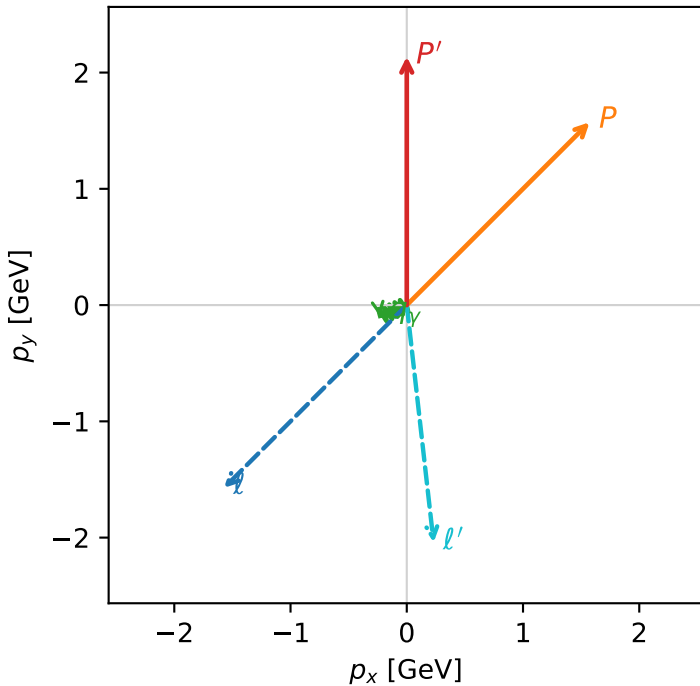
w_purity_low_egamma_ty_lepton_region_3 P_W=0.155334
 region: low_Egamma_Ty_lepton_region_3
 outgoing-state purity: 0.5
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell_0},h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.13719$
 $\theta_{in}=1.57079$, $\phi_l=3.92699$, $\phi_P=0.785398$
 $E_\gamma=0.25$, $\phi_\gamma=3.53429$
 $Q^2=3.4447$, $x_B=0.68611$, $t=-2.78602$, $W^2=2.45576$, $y=0.243294$
 $\theta(\ell',\gamma)=73.9548$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.2244$ $p_x= -1.5729$ $p_y= -1.5729$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 1.5729$ $p_y= 1.5729$ $p_z= 0.0000$
 ℓ' $E= 2.0545$ $p_x= 0.2310$ $p_y= -2.0415$ $p_z= 0.0000$
 P' $E= 2.3340$ $p_x= 0.0000$ $p_y= 2.1372$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2310$ $p_y= -0.0957$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=8, retained=76.8%)

